Lab - Asking the Right Question

Objectives

This lab provides opportunities for learners to begin a personal analytic project.

- Choose a topic to investigate.
- Formulate a question to answer with the analysis.
- Determine the data elements that are needed to find the answers.

Background / Scenario

In this lab, you will begin your first analytic project, starting with the “Ask the Question” step of the process. Choose a topic that you are interested in and that you want to see through to the end. In most data analysis projects, there is an identified need or benefit that informs the question that starts the process. The question determines what type of analysis and what data elements are necessary to produce the desired outcomes.

Once you have formulated a question, you can make hypothesis as to the outcome of the analysis. After completing your analysis, later in the course, you can compare your initial hypothesis with the actual outcome.

Required Resources

- A device with an internet browser and an internet connection.

Instructions

Part 1: Determine a Topic

Did you ever wonder about something that you were unable to find an answer for, or that you were skeptical of the answers that you did find? Topics that you want to know more about are excellent choices for your first analytic project.

Step 1: Ask a question

It could be something as simple as “does having Batman appear in your movie make it more successful at the box office?” or “which flavors of ice cream are most popular with different age groups?”. Or it could be something more complex, like “what cryptocurrency is the best investment based on past year performance?”.

a. Write down your question. What is the benefit of determining the answer to this question? In most data analytic projects, the potential benefit is the reason for the project. For example, a marketing executive would benefit from knowing which age groups prefer certain flavors of ice cream, because it could help determine where to place advertising, or which flavors to stock in towns with colleges or universities.
b. Review the four different types of data analysis described in the module. What type of analysis do you think fits best to uncover the answer to your question? In most cases, descriptive analysis is a good choice if the answer will be determined by analyzing historical data.

Step 2: Determine the Data Needed

Now that you have a question in mind, think about the data elements that you might need to analyze in order to come up with an answer to the question.

a. Make a list of the data elements that you identify. An example: If the question is “does having Batman appear in your movie make it more successful at the box office?”, some of the needed data elements would be:
    - Movies with Batman as a character
    - Successful movies made in the same timeframe
    - Box office sales and revenue of movies in the same timeframe
b. Open a web browser and use any search engine to search for the data elements that you have identified.
c. List sources that you found that may have data relating to the elements you identified.
d. You may also want to investigate sources, such as Kaggle, that provide sample data for analysis.

Reflection Questions

1. Why is it important to identify the question that needs to be answered by the analysis before beginning the project?
2. Name some sources of open data for analysis that you found while searching for your data elements.

Study Time Preferences of Students

Step 1: Ask a Question

Which study time is preferred by students: morning, afternoon, or night?

The Benefits

01 Students plan their schedules more effectively

02 Teachers decide when to assign activities or deadlines

03 Students manage their time better and reduce stress

Descriptive Analysis

Step 2: Determine the Data Needed

- Student identifier
- Preferred study time (Morning / Afternoon / Night)
- Number of students choosing each study time

Student	Study hours per week	Preferred Study Time	Method of Study
	13	MORNING	Individual
	14	MORNING	Individual
	7	EVENING	Individual
	21	AFTERNOON	Group
	7	EVENING	Individual
	20	MORNING	Group
	12	EVENING	Individual
	12	MORNING	Individual
	7	AFTERNOON	Group
	21	MORNING	Individual

Study Habits Questions	Chronotype	Mean	Interpretation
I prefer study in quite place instead of place with background sounds.	Early Bird	3.95	Neutral
	Night Owl	3.79	Neutral
I prefer to learn in daytime instead of at night.	Early Bird	3.60	Neutral
	Night Owl	2.35	Partly Disagree
I prefer to study alone instead of learning in group.	Early Bird	3.92	Neutral
	Night Owl	3.57	Neutral
I will take notes when learning.	Early Bird	3.99	Neutral
	Night Owl	3.66	Neutral
I prefer using electronic device to take note instead of notebook.	Early Bird	2.72	Neutral
	Night Owl	2.97	Neutral
I will review my book after lessons.	Early Bird	3.33	Neutral
	Night Owl	2.73	Neutral
I will preview my book before class.	Early Bird	2.78	Neutral
	Night Owl	2.31	Partly Disagree

Table 5. Early Bird and Night Owl study Habits Comparison

SOURCES:

- https://papers.iafor.org/wp-content/uploads/papers/iicah2023/IICAH2023_66769.pdf
- https://www.studocu.com/ph/document/technological-institute-of-the-philippines/statistical-analysis-with-software-application/p2-survey-and-data-collection-study-habits/105638262

Reflection Questions

1) Identifying the question defines the goal of the analysis, determines what data is needed, prevents the collection of unnecessary or irrelevant data, and guides the type of analysis to be performed

2) Kaggle, Google Dataset Search, GitHub